

Urinary Tract Infection in Pregnancy

Lecture

Week 3 · Medical disorders & PPH

Obstetrics

Dr Nanzira

Progesterone-induced ureteric relaxation and compression by the gravid uterus cause urinary stasis, so bacteriuria in pregnancy readily ascends to the kidney. Screen for and treat asymptomatic bacteriuria, treat cystitis with a pregnancy-safe antibiotic (nitrofurantoin, amoxicillin if sensitive, cephalexin) for 5–7 days, and admit every case of pyelonephritis for IV ceftriaxone, sepsis screening and fetal monitoring.

Must know before the ward round

1. Pregnancy predisposes to UTI through progesterone (smooth-muscle relaxation, ureteric dilatation, reduced peristalsis), mechanical compression (right more than left), vesicoureteric reflux, glycosuria and a larger, slower-emptying bladder.
2. Asymptomatic bacteriuria (ASB, 10^5 CFU/mL or more of one organism on a clean-catch MSU without symptoms) affects about 2–10% of pregnant women (about 13% in a Mulago study); untreated, up to 30–40% progress to pyelonephritis, so screen and treat for 7 days.
3. Dipstick nitrites are specific but only about 50% sensitive, so a negative dipstick does not exclude ASB. Send a clean-catch MSU for culture before the first antibiotic dose wherever possible.
4. Cystitis (dysuria, frequency, suprapubic pain, no fever) is treated as an outpatient for 5–7 days. Fever or loin pain means pyelonephritis, and every case is admitted.
5. *E. coli* causes 70–80% of UTIs; Ugandan isolates are often resistant to co-trimoxazole and amoxicillin, so culture where possible and do a test of cure.
6. First-line oral options are nitrofurantoin 100 mg MR 12-hourly, cephalexin 500 mg 8–12-hourly, or amoxicillin 500 mg 8-hourly only if sensitive, for 5–7 days, with a test-of-cure culture 1–2 weeks after treatment.
7. Safe drugs are nitrofurantoin (avoid near term and in G6PD deficiency), amoxicillin or amoxicillin-clavulanate, cephalexin, cefixime and ceftriaxone. Avoid fluoroquinolones and tetracyclines, trimethoprim in the first trimester, and sulfonamides near term.
8. Every case of pyelonephritis in pregnancy is admitted – cultures, IV ceftriaxone 1–2 g daily, careful fluids, paracetamol, fetal monitoring, then oral step-down to complete 10–14 days.
9. Screen every febrile pregnant woman for sepsis (temperature above 38 or below 36 °C, HR above 100, RR 22 or more, SBP 100 or below, confusion, SpO₂ below 95%, lactate above 2) and do an mRDT; start the sepsis bundle within 1 hour.
10. Fever persisting after 48–72 hours of IV antibiotics means check the culture (ESBL?), do a renal ultrasound for obstruction or abscess, and rethink the diagnosis.
11. Complications of pyelonephritis are sepsis/septic shock, ARDS (endotoxin capillary leak, worsened by fluid overload and beta-agonists), anaemia, AKI, preterm labour and recurrence.
12. Recurrent UTI or pyelonephritis needs nightly prophylaxis (nitrofurantoin 50–100 mg or cephalexin 250–500 mg) until after delivery, plus a search for a cause (diabetes, stones, sickle cell, HIV, anatomical anomaly).

1 In a nutshell

2 Definitions & classification

3 Epidemiology & Uganda context

4 Pathophysiology

- 5 Clinical acumen
- 6 Clinical workup
- 7 Differential diagnosis
- 8 Investigations
- 9 Management
- 10 Complications
- 11 Ward-round & exam pearls
- 12 Mnemonics & memory aids
- 13 High-yield summary

In a nutshell

A **urinary tract infection (UTI)** in pregnancy ranges from **asymptomatic bacteriuria (ASB)**, bacteria growing silently in the bladder, through **acute cystitis** (bladder infection with symptoms) to **acute pyelonephritis** (kidney infection with fever and loin pain).

Pregnancy **does not make bacteriuria more common**, but it makes bacteriuria **far more dangerous**: the dilated, sluggish ureters let bacteria climb to the kidney. That is why a condition we usually ignore in non-pregnant women (ASB) is **screened for and treated** in pregnancy.

Why it matters in Uganda: pyelonephritis is one of the commonest reasons for antenatal admission, a common cause of **fever in pregnancy** (alongside malaria), an important trigger of **preterm labour**, and a cause of **maternal sepsis**, a leading direct cause of maternal death. Antibiotic resistance is high, and urine culture is not available at most lower-level facilities.

CLINICAL PEARL

The one thing never to forget: **fever plus loin pain in a pregnant woman is pyelonephritis until proven otherwise, and every case is admitted**. Pyelonephritis in pregnancy is a sepsis risk, not an outpatient illness.

Definitions & classification

Term	Definition
Significant bacteriuria	10⁵ colony-forming units (CFU)/mL or more of a single uropathogen on a clean-catch midstream urine (MSU) culture (lower counts count on a catheter specimen)
Asymptomatic bacteriuria (ASB)	Significant bacteriuria without symptoms
Acute cystitis (lower UTI)	Bladder infection: dysuria, frequency, urgency, suprapubic pain, haematuria; no fever or loin pain
Acute pyelonephritis (upper UTI)	Infection of the renal pelvis and parenchyma: fever, rigors, loin (flank) pain, renal angle tenderness , nausea/vomiting, ± lower urinary symptoms
Recurrent UTI in pregnancy	Two or more UTIs in pregnancy (or one UTI plus ASB), or relapse after treatment
Relapse vs reinfection	Relapse = same organism within 2 weeks (treatment failure, possible structural cause); reinfection = new organism later
Group B streptococcal (GBS) bacteriuria	GBS in urine at any colony count : a marker of heavy genital colonisation; treat and give intrapartum antibiotic prophylaxis
Urosepsis	Life-threatening organ dysfunction due to a dysregulated response to a urinary infection

Epidemiology & Uganda context

- **ASB:** about **2–10%** of pregnant women worldwide. A study at **Mulago Hospital** found ASB in about **13%** of antenatal mothers, with *E. coli* the commonest isolate and high resistance to commonly used oral antibiotics.
- Untreated ASB progresses to **pyelonephritis in 20–40%** of pregnant women (versus about 1–2% if treated). Treating ASB reduces pyelonephritis by about three-quarters and is associated with fewer low-birth-weight babies.
- **Pyelonephritis** complicates about **1–2% of pregnancies**, mostly in the **second and third trimesters**, and is **more often right-sided**.
- **Organisms:** *Escherichia coli* (70–80%), *Klebsiella*, *Proteus*, *Enterobacter*, *Enterococcus*, **group B streptococcus**, *Staphylococcus saprophyticus*.
- **Resistance in Uganda:** many community *E. coli* isolates are resistant to **co-trimoxazole**, **ampicillin/amoxicillin** and **tetracyclines**; **extended-spectrum beta-lactamase (ESBL)** producers are increasingly reported, including resistance to ceftriaxone. Nitrofurantoin generally retains good activity against *E. coli*.
- **Facility realities:** dipstick urinalysis is available at most HC IIIs/IVs (if strips are in stock); microscopy at HC IV; **culture and sensitivity** mainly at regional and national referral hospitals and private labs, with results in 2–3 days. **G6PD deficiency** is relatively common in Uganda (relevant to nitrofurantoin).
- **Risk factors:** previous UTI, diabetes (glycosuria), **sickle cell disease and trait**, **HIV**, urinary tract anomalies or stones, low socio-economic status, sexual activity, multiparity, poor hygiene/water access, catheterisation.

UGANDA CONTEXT

WHO ANC (2016) recommends **screening for ASB** in pregnancy. Where culture is not available, WHO suggests **midstream urine Gram staining** over dipstick testing, but in most Ugandan HC IIIs the practical screen is a **urine dipstick** (nitrites and leucocyte esterase) at booking and when symptomatic. Remember: **a negative dipstick does not exclude ASB** (nitrite sensitivity is only about 50%). Send urine for culture wherever you can, and always for pyelonephritis.

Pathophysiology

Why pregnancy predisposes to ascending infection

Change	Mechanism	Effect
Progesterone	Relaxes smooth muscle of the ureters and bladder	Ureteric dilatation , reduced peristalsis, urinary stasis ; larger, floppier bladder that empties incompletely
Mechanical compression by the enlarging uterus (from about 20 weeks) at the pelvic brim	Ureters compressed where they cross the iliac vessels	Physiological hydroureter and hydronephrosis (up to 90% of women by the third trimester)
Right more than left	The uterus is dextrorotated ; the sigmoid colon cushions the left ureter ; the right ovarian vein crosses the right ureter	Right-sided pyelonephritis is more common
Vesicoureteric reflux	Bladder oedema/hyperaemia and displaced trigone reduce the valve effect at the ureteric orifices	Bacteria reflux up the ureters
Urine composition	Glycosuria (lower renal threshold), aminoaciduria , more alkaline urine	Better culture medium for bacteria

Change	Mechanism	Effect
Increased GFR and urine volume	Dilated system holds a larger volume of stagnant urine (up to 200 mL)	Stasis
Immune modulation	Relative suppression of cell-mediated immunity	Infection spreads more easily
Short female urethra near the vagina and anus	Colonisation of the periurethral area by gut flora	Entry of organisms (same as non-pregnant)

```

Gut flora (E. coli) colonise perineum -> urethra -> bladder (ASB / cystitis)
      |
progesterone + uterine compression: dilated, sluggish ureters, reflux
      |
ascent to the renal pelvis and parenchyma
      |
ACUTE PYELONEPHRITIS
      |
endotoxin (lipopolysaccharide) -> cytokines -> capillary leak
      |           |           |           |
SEPSIS/SHOCK   ARDS (lung leak) HAEMOLYSIS/ANAEMIA + AKI PROSTAGLANDINS
                                     -> PRETERM LABOUR
  
```

Why pyelonephritis is so dangerous in pregnancy

- **Endotoxin** from Gram-negative bacteria triggers a strong cytokine response. In pregnancy, **low plasma oncotic pressure** (dilutional hypoalbuminaemia) and high plasma volume make the lungs prone to **capillary leak: ARDS/pulmonary oedema**, made worse by **aggressive IV fluids** and **beta-agonist tocolytics**.
- **Endotoxin damages red cell membranes** (haemolysis), causing **anaemia** in about a quarter of women.
- **Transient renal dysfunction** (fall in creatinine clearance) is common.
- **Inflammation stimulates prostaglandins and uterine contractions**: preterm labour. **Fever** itself causes fetal tachycardia and increases fetal oxygen demand.
- **ASB itself** is linked to preterm birth and low birth weight, probably through the same inflammatory pathways.

Clinical acumen

Focused history

Question	Why it matters
Burning on passing urine? Is it internal (in the urethra) or external (as urine touches sore vulval skin)?	Internal dysuria suggests UTI; external suggests vulvovaginitis (candida, trichomonas, herpes)
Frequency, urgency, suprapubic pain, blood in urine, smelly urine?	Cystitis. Frequency alone is normal in pregnancy : ask whether it is new or different
Fever, rigors, loin/back pain, vomiting?	Pyelonephritis; vomiting means oral treatment will fail
Abdominal tightening, pain, vaginal bleeding, leaking liquor, fetal movements?	Preterm labour , abruption, PPROM/chorioamnionitis
Headache, cough, breathlessness?	Malaria, pneumonia; breathlessness during treatment suggests ARDS/fluid overload
Vaginal discharge, itching, new partner?	STI (chlamydia/gonorrhoea cause dysuria with sterile pyuria), vaginitis

Question	Why it matters
Previous UTIs, pyelonephritis, kidney stones, known urinary tract abnormality?	Recurrence risk; need for prophylaxis/imaging
Diabetes, sickle cell disease/trait , HIV (and co-trimoxazole use), TB	Risk factors; atypical organisms; drug choices
Drug allergies (penicillin), G6PD deficiency , recent antibiotics	Choice of antibiotic; resistance
Gestational age	Drug choice (trimethoprim first trimester, nitrofurantoin near term), steroids if preterm labour
Malaria test done? Travel, ITN use	In Uganda every febrile pregnant woman needs an mRDT/blood smear

Examination

- **Vital signs are the priority:** temperature, pulse, **respiratory rate**, BP, SpO₂, level of consciousness, urine output. Use the **modified obstetric early warning score (MEOWS)** chart. Tachycardia above 100, RR 22 or more, SBP 100 or below, or confusion suggest **sepsis**.
- **General:** hydration, pallor (anaemia), jaundice, rash.
- **Chest:** crackles (pneumonia, pulmonary oedema/ARDS).
- **Abdomen:** suprapubic tenderness (cystitis); **renal angle (costovertebral angle) tenderness** (pyelonephritis; percuss gently); right iliac fossa/flank tenderness (appendicitis: the appendix moves upwards and laterally in pregnancy); RUQ (cholecystitis, HELLP).
- **Obstetric:** SFH, **uterine tenderness** (chorioamnionitis, abruption, red degeneration of a fibroid), **contractions**, fetal heart rate (**fetal tachycardia** with maternal fever).
- **Speculum** if discharge, bleeding or leaking (PPROM, cervicitis); cervical assessment if contractions.

Red flags

RED FLAG

- **Signs of sepsis:** temperature above 38 °C or below 36 °C, HR above 100 (persistently above 110), RR 22 or more, SBP 100 mmHg or below, confusion, SpO₂ below 95%, oliguria (below 0.5 mL/kg/h), mottled skin, raised lactate. Start the **sepsis bundle within 1 hour**.
- **Breathlessness, falling SpO₂ or crackles** during treatment: **ARDS or pulmonary oedema**; stop IV fluids, oxygen, senior review.
- **Persistent fever after 48–72 hours** of appropriate IV antibiotics: resistant organism, **obstruction (stone)**, renal or perinephric abscess, or wrong diagnosis.
- **Regular contractions or cervical change:** preterm labour.
- **Reduced fetal movements, fetal tachycardia above 160 or bradycardia.**
- **Rising creatinine / oliguria:** AKI.
- **Vomiting:** cannot take oral treatment; needs IV therapy.

Thinking like a clinician

- **In Uganda, fever in pregnancy is malaria AND something else until proven otherwise.** Always do an mRDT, but a positive mRDT does not exclude pyelonephritis (both can co-exist; the mRDT can stay positive for weeks after treated malaria).
- **Distinguish cystitis from pyelonephritis clinically: fever and loin pain = upper tract = admit.** Cystitis does not cause high fever.

- **Pyuria is not bacteriuria.** Leucocytes without nitrites and a negative culture (sterile pyuria) suggest **chlamydia/gonorrhoea, TB of the urinary tract, contamination with vaginal discharge, or recent antibiotics.**
- **Contaminated samples** are common: many squamous epithelial cells and mixed growth mean the MSU was not clean. Repeat it.
- **Right loin pain has a big differential** in pregnancy: pyelonephritis, appendicitis, renal colic, cholecystitis, HELLP, abruption, red degeneration of a fibroid, ovarian torsion. Examine the uterus and check urine, BP and bloods.
- **When the patient does not improve**, do not simply add more antibiotics: check the culture result, image the kidneys (ultrasound for obstruction/abscess), reconsider the diagnosis (malaria, chorioamnionitis, appendicitis, pneumonia).
- **Over-hydration is a real danger:** pregnant women with pyelonephritis are prone to pulmonary oedema. Give fluids in small boluses with reassessment and keep a fluid balance chart.
- **Common student mistakes:**
 1. Prescribing **ciprofloxacin** (commonly used for UTI in non-pregnant patients in Uganda) to a pregnant woman.
 2. Treating ASB with a **single dose** or 3 days by default (use 5–7 days; WHO recommends 7 days for ASB).
 3. Forgetting a **test of cure** urine culture 1–2 weeks after treatment.
 4. Sending a woman with pyelonephritis home on oral antibiotics.
 5. Using **nitrofurantoin for pyelonephritis** (it does not achieve adequate renal tissue or blood levels).
 6. Forgetting the **fetus** (fetal heart, contractions, steroids if preterm).

Clinical workup

On an antenatal ward the question is almost always one of three: **Is this bacteriuria at all? Is it lower (cystitis) or upper (pyelonephritis)? Is she septic?** The steps below answer them in that order, while keeping an eye on the fetus and on the other causes of fever in Uganda.

Step 0: First 5 minutes

1. **Full set of vital signs on a MEOWS chart:** temperature, pulse, RR, BP, SpO₂, level of consciousness, and the time she last passed urine.
2. **Sepsis screen:** suspected infection plus **any one** of temperature **above 38 °C or below 36 °C**, HR **above 100**, RR **22 or more**, SBP **100 mmHg or below**, new confusion, SpO₂ **below 95%**, or oliguria. If positive: **call a senior** and start the **sepsis bundle within 1 hour** (oxygen, blood cultures, IV antibiotics, fluids, lactate and Hb, urine output).
3. **IV access and bloods at the same time:** FBC, **blood cultures**, creatinine, **mRDT**, glucose, lactate where available.
4. **Urine sample before the first antibiotic dose:** clean-catch MSU (or catheter specimen if she is too ill), for dipstick and **culture**.
5. **Start treatment without waiting for results:** **ceftriaxone 1 g IV daily (2 g if septic)**, paracetamol 1 g, and **500 mL crystalloid boluses** only if hypotensive, reassessing the chest and SpO₂ after each.
6. **Fetal heart and uterus:** FHR (tachycardia above 160 is common with maternal fever), contractions, uterine tenderness.

RED FLAG

Breathless, SpO₂ falling or crackles after IV fluids = pulmonary oedema/ARDS until proven otherwise. Stop the fluids, sit her up, give oxygen and call for help.

Step 1: Focused history

- ☐ **Urinary symptoms:** dysuria (internal or external?), new frequency, urgency, suprapubic pain, haematuria, smelly urine. **Frequency alone is normal in pregnancy;** external dysuria suggests vulvovaginitis.
- ☐ **Upper tract symptoms: fever, rigors, loin/back pain, nausea and vomiting.** Any of these = pyelonephritis; vomiting means oral drugs will fail.
- ☐ **Gestational age:** decides drug choice (no trimethoprim in the first trimester; no nitrofurantoin or sulfonamides near term) and steroids if preterm labour threatens.
- ☐ **Obstetric symptoms:** tightenings, bleeding, **leaking liquor**, fetal movements. Preterm labour, abruption, PPROM/chorioamnionitis.
- ☐ **Other fever sources:** headache, myalgia (malaria), cough (pneumonia), offensive discharge (chorioamnionitis), RIF pain (appendicitis).
- ☐ **Previous UTIs this pregnancy:** treated with what, and was a **test of cure** done? Recurrence or relapse changes the plan (culture, prophylaxis, imaging).
- ☐ **Risk factors:** diabetes, **sickle cell disease or trait**, **HIV** (and co-trimoxazole), stones, known urinary tract anomaly, recent catheter.
- ☐ **Antibiotics already taken** (including pharmacy-bought ciprofloxacin): affects culture yield and resistance.
- ☐ **Allergies** (penicillin: type of reaction) and **G6PD deficiency** (nitrofurantoin).
- ☐ **Vaginal discharge, itch, new partner:** STI or vaginitis causing dysuria and sterile pyuria.

Step 2: Focused examination

Area	Look for	Key positives and negatives to document
General	Toxic look, rigors, hydration, pallor (haemolysis), jaundice, confusion	"Febrile, flushed, dry mucosae, not pale, alert"
Vital signs	MEOWS triggers	"T 39.2 °C, HR 118, RR 22, BP 104/66, SpO ₂ 97% on air: sepsis screen positive"
Chest	Crackles (pneumonia; later, pulmonary oedema/ARDS)	"Chest clear" (re-examine after fluids)
Abdomen	Suprapubic tenderness (cystitis); renal angle tenderness on gentle percussion (pyelonephritis); RIF/flank tenderness (appendicitis); RUQ (cholecystitis, HELLP)	"Right renal angle tenderness; no RIF tenderness; bladder not palpable"
Obstetric	SFH, uterine tenderness (chorioamnionitis, abruption, red degeneration), contractions , FHR	"Uterus soft, non-tender, no contractions, FHR 170"
Speculum (if discharge, bleeding or leaking)	Liquor (PPROM), cervicitis, discharge	"No liquor, no discharge"
Vaginal examination	Only if contractions or suspected labour (and not if PPROM without labour)	Cervical change = preterm labour

Step 3: Bedside tests

Test	How to read it
Urine dipstick (fresh sample)	Nitrites positive: Gram-negative bacteria (specific; only about 50% sensitive). Leucocyte esterase: pyuria. Both positive with symptoms = treat as UTI. LE alone = possible contamination, STI or vaginitis. Protein: UTI or pre-eclampsia (check BP). Glucose: lower renal threshold or diabetes. Ketones: vomiting/starvation. Blood: infection, stone, contamination

Test	How to read it
Look at the urine	Cloudy, offensive urine supports infection; clear urine makes it less likely (not excluded)
mRDT (± blood smear)	Every febrile woman. A positive HRP2 mRDT can stay positive for weeks after treated malaria and does not exclude pyelonephritis
RBS	Undiagnosed diabetes (recurrent UTI) or stress hyperglycaemia
HemoCue Hb	Haemolytic anaemia from endotoxin
HIV rapid test (if status unknown)	Risk factor; changes prophylaxis (co-trimoxazole)
Pulse oximetry	Baseline and after every fluid bolus (ARDS watch)
Hourly urine output (catheter if septic)	Aim 0.5 mL/kg/h or more
Fetal heart / CTG (viable gestation)	Tachycardia settles as maternal fever falls; persistent abnormality needs review
Point-of-care ultrasound	Fetal viability, liquor, placenta; gross hydronephrosis if trained (formal renal scan still needed)

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Urine microscopy	Symptoms, positive dipstick	Pus cells more than 10 per high-power field , bacteria; white cell casts = pyelonephritis ; many squamous cells = contaminated sample	Supports diagnosis; repeat a contaminated sample. HC IV and above
Urine culture and sensitivity (MSU before antibiotics)	Booking screen (ASB), every symptomatic UTI where possible, always in pyelonephritis , recurrence, treatment failure	10⁵ CFU/mL or more of one organism (lower counts on a catheter specimen); GBS at any count ; mixed growth = contamination	Targets therapy (resistance to amoxicillin/co-trimoxazole, ESBL); GBS → intrapartum prophylaxis. Regional/national referral and private labs; results in 2–3 days
Test-of-cure culture	1–2 weeks after finishing treatment	Persistent bacteriuria in about 1 in 5	Re-treat by sensitivity; consider prophylaxis
Blood cultures	Pyelonephritis, sepsis, before antibiotics	Positive in 15–20% of pyelonephritis	Guides antibiotics; bacteraemia = closer monitoring. Regional/national referral
FBC	Pyelonephritis, sepsis	WBC raised; anaemia (haemolysis, about a quarter); low platelets (sepsis/DIC)	Severity; transfusion; DIC workup
Creatinine, urea, electrolytes	Pyelonephritis, sepsis, before gentamicin	Creatinine above ~77–80 µmol/L is suspicious in pregnancy (normal 35–70)	AKI: adjust or avoid gentamicin; fluid balance

Test	When	Expected / abnormal finding	How it changes management
Lactate (where available)	Sepsis screen positive	Above 2 mmol/L = hypoperfusion; 4 or more = severe	Fluid boluses, HDU/ICU; lactate above 2 alone meets the sepsis trigger
LFTs	RUQ pain, jaundice, hypertension	Raised ALT, bilirubin	HELLP, cholecystitis, hepatitis
Renal ultrasound	No response in 48–72 h , recurrent pyelonephritis, haematuria or colic, suspected stone	Hydronephrosis out of proportion to gestation, stone, renal/perinephric abscess	Urology: ureteric stent or nephrostomy; drainage. Hospitals
Obstetric ultrasound	Pyelonephritis, contractions, uncertain dates	GA, liquor, placenta, fetal wellbeing	Steroids, timing, differential (abruption, fibroid degeneration)
Chest X-ray (shielded)	Breathless, SpO ₂ below 95%, crackles	Bilateral infiltrates (ARDS/oedema), lobar consolidation	Stop fluids, oxygen, diuretic if oedema, ICU
Gram stain of MSU	Where culture is unavailable (WHO-suggested ASB screen)	Bacteria on Gram stain	Treat ASB empirically, then test of cure if possible

Step 5: Putting it together

Model one-line summary

"A 24-year-old G2P1 at 26 weeks with 2 days of fever, rigors, right loin pain and vomiting, febrile 39.2 °C, tachycardic 118, RR 22, BP 104/66, with right renal angle tenderness, nitrite-positive urine, negative mRDT and fetal tachycardia."

Problem list

1. **Acute right pyelonephritis with sepsis** (sepsis screen positive).
2. Vomiting and dehydration: needs IV therapy, but at risk of fluid overload/ARDS.
3. **Fetal tachycardia** and risk of **preterm labour** at 26 weeks.
4. Previous untreated cystitis at 14 weeks without test of cure: recurrence risk, needs prophylaxis.

Working diagnosis: acute pyelonephritis in pregnancy complicated by sepsis, probably *E. coli*.

Worked example 1: reading urine results

Woman	Symptoms	Dipstick	Microscopy / culture	Interpretation and action
A (booking, 14 weeks)	None	Nitrites negative , LE negative	Culture: 10⁵ CFU/mL <i>E. coli</i> , sensitive to nitrofurantoin and cephalexin	ASB (a negative dipstick did not exclude it). Nitrofurantoin 100 mg MR 12-hourly for 7 days ; test of cure in 1–2 weeks
B (22 weeks)	Dysuria, frequency, no fever	Nitrites + , LE ++	Pus cells 25/HPF; culture pending	Acute cystitis . Send MSU first, start cephalexin 500 mg 8-hourly for 5–7 days ; adjust to sensitivity; safety-net advice
C (20 weeks)	Dysuria, itch, discharge	Nitrites negative, LE +	Pus cells 15/HPF, many squamous cells; culture: mixed growth	Contaminated sample and likely vulvovaginitis . Speculum, treat the discharge, repeat a proper clean-catch MSU
D (30 weeks)	None	Negative	Culture: GBS 10⁴ CFU/mL	GBS bacteriuria (any count counts): treat now and give intrapartum IV benzylpenicillin in labour

Woman	Symptoms	Dipstick	Microscopy / culture	Interpretation and action
E (18 weeks)	Dysuria, weight loss, HIV-positive	LE ++, nitrites negative	Pus cells 30/HPF, culture sterile twice	Sterile pyuria: think chlamydia/gonorrhoea, recent antibiotics, urinary TB (early-morning urine for GeneXpert)

Worked example 2: the sepsis screen in pyelonephritis

Admission observations for woman above (the one-line summary):

Parameter	Value	Trigger?
Temperature	39.2 °C	Yes (above 38)
Heart rate	118/min	Yes (above 100)
Respiratory rate	22/min	Yes (22 or more)
Systolic BP	104 mmHg	No (but close to 100)
SpO ₂	97%	No
Consciousness	Alert	No
Lactate	2.6 mmol/L	Yes (above 2)
WBC / Hb / creatinine	17 × 10 ⁹ /L / 10.1 g/dL / 88 µmol/L	Leucocytosis; creatinine raised for pregnancy (AKI risk)

Interpretation: sepsis from a urinary source (several triggers plus raised lactate), not yet septic shock.

Action within 1 hour: senior informed; oxygen only if SpO₂ falls below 94%; **blood and urine cultures**, then **ceftriaxone 2 g IV** (avoid gentamicin for now given the creatinine); **500 mL Ringer's lactate** boluses with chest and SpO₂ checks after each; catheter and hourly urine output; paracetamol 1 g; repeat lactate. FBC and creatinine daily. CTG: expect the fetal tachycardia to settle as the fever falls. **Review at 48–72 hours:** if still febrile, check culture (ESBL?) and arrange a **renal ultrasound**.

Differential diagnosis

Condition	Distinguishing features	Key investigation
Acute cystitis	Dysuria, frequency, urgency, suprapubic pain; afebrile	Dipstick (nitrites, LE), MSU culture
Acute pyelonephritis	Fever, rigors, loin pain, renal angle tenderness, vomiting	Urinalysis, culture, FBC, blood culture, creatinine
Malaria	Fever, rigors, headache, myalgia; no urinary symptoms (but can co-exist)	mRDT / blood smear
Vulvovaginal candidiasis / trichomoniasis	External dysuria, itch, discharge	Speculum, wet mount
Urethritis / cervicitis (chlamydia, gonorrhoea)	Dysuria, discharge, sterile pyuria	NAAT where available, syndromic management
Preterm labour	Regular painful contractions, cervical change; may be triggered by UTI	Speculum/VE, CTG, urine culture
Chorioamnionitis	Fever, uterine tenderness , offensive liquor, fetal tachycardia, often after PROM	Clinical, FBC, HVS
Placental abruption	Constant pain, tender hard uterus, bleeding, fetal distress	Clinical, CTG, ultrasound

Condition	Distinguishing features	Key investigation
Appendicitis	Right-sided pain (higher in pregnancy), anorexia, low-grade fever, guarding; urinary symptoms can occur if inflamed appendix lies near the ureter	Clinical, ultrasound
Renal colic / urolithiasis	Severe colicky loin-to-groin pain, haematuria, little fever unless infected obstruction	Renal ultrasound (distinguish from physiological hydronephrosis)
Pre-eclampsia / HELLP	RUQ/epigastric pain, hypertension, proteinuria	BP, urine protein, FBC (platelets), LFTs
Red degeneration of fibroid	Localised tender mass on the uterus, mid-pregnancy, low-grade fever	Ultrasound
Pneumonia (lower lobe)	Cough, pleuritic pain, crackles; pain can refer to the loin	CXR (shielded)
Urinary TB	Chronic symptoms, sterile pyuria, HIV-positive, weight loss	Early-morning urine for GeneXpert/AFB

Investigations

Each test, with its interpretation and Ugandan availability, is in **Clinical workup, Steps 3 and 4** above. The sample itself is where most mistakes happen, so learn to teach this.

How to collect a clean-catch MSU (teach the patient)

1. Wash hands; separate the labia with one hand.
2. Clean the vulva **front to back** with water (not antiseptic, which can kill bacteria in the sample).
3. Pass the **first part of the stream into the toilet**, then catch the **middle part** in a sterile container without touching the inside.
4. Deliver to the lab **within 2 hours** (or refrigerate at 4 °C for up to 24 hours).
5. Label the form with **gestation, symptoms and any antibiotics already taken**, and whether it is a catheter specimen (lower counts are significant).

Management

1. Screening and treating asymptomatic bacteriuria

1. **Screen at booking** (ideally 12–16 weeks): urine culture where available; otherwise Gram stain or dipstick (and culture if dipstick positive).
2. **If significant bacteriuria**: treat according to sensitivities for **5–7 days** (WHO: a **7-day** regimen for ASB).
3. **Test of cure**: repeat culture **1–2 weeks after** completing treatment, then **each trimester** (or monthly in high-risk women).
4. **Two episodes** (or one episode that relapses): consider **suppressive prophylaxis** for the rest of pregnancy.
5. **GBS in urine (any count)**: treat now and give **intrapartum IV antibiotic prophylaxis** in labour (benzylpenicillin 3 g IV loading, then 1.8 g IV 4-hourly until delivery; clindamycin 900 mg IV 8-hourly if allergic).

2. Acute cystitis (outpatient, HC III upwards)

1. Confirm: symptoms + dipstick (nitrites/LE); **send MSU for culture before the first dose** where possible.
2. Start **empirical oral antibiotic** (table), adjust to sensitivities when available.

3. **Advice:** drink plenty of fluids (2–3 L/day), void regularly and after intercourse, wipe front to back, paracetamol for discomfort.
4. **Return immediately** if fever, loin pain, vomiting, contractions or bleeding.
5. **Test of cure** 1–2 weeks after treatment.

Oral antibiotics for ASB and cystitis in pregnancy

Drug	Dose	Route	Frequency	Duration	Notes
Nitrofurantoin	100 mg (modified-release) or 50–100 mg (standard)	Oral	12-hourly (MR) or 6-hourly (standard), with food	5–7 days	Good activity against <i>E. coli</i> ; not for pyelonephritis; avoid at term (from about 36–38 weeks), in labour, and in maternal G6PD deficiency (haemolysis in mother or neonate); avoid if eGFR below 45
Amoxicillin	500 mg	Oral	8-hourly	7 days	Only if the organism is sensitive (high resistance in Uganda)
Amoxicillin-clavulanate	625 mg (500/125)	Oral	8-hourly	5–7 days	Broader cover; useful empirical choice
Cephalexin (cefalexin)	500 mg	Oral	8–12-hourly (up to 6-hourly)	5–7 days	Safe throughout pregnancy
Cefixime	200 mg (or 400 mg once daily)	Oral	12-hourly	5–7 days	Oral third-generation cephalosporin; good step-down option
Fosfomicin (where available)	3 g	Oral	Single dose	Once	Safe; limited availability in Uganda
Penicillin allergy	Nitrofurantoin or (if mild non-anaphylactic allergy) cephalexin				Avoid cephalosporins after anaphylaxis to penicillin

DRUG DOSING

Antibiotics to avoid in pregnancy

- **Fluoroquinolones** (ciprofloxacin, levofloxacin): cartilage/joint damage in animal studies; avoid (use only if no alternative, on specialist advice).
- **Tetracyclines** (doxycycline): **tooth discolouration and impaired fetal bone growth**; maternal hepatotoxicity.
- **Trimethoprim**: a **folate antagonist**; avoid in the **first trimester** (neural tube and other defects).
- **Sulfonamides / co-trimoxazole** (as UTI treatment): avoid in the **first trimester** and **near term** (displaces bilirubin from albumin: **neonatal jaundice/kernicterus**; haemolysis in G6PD deficiency). HIV-positive women on **co-trimoxazole prophylaxis** continue it (the benefit outweighs the risk), but it should not be relied on as UTI treatment.
- **Aminoglycosides** (gentamicin): **fetal ototoxicity/nephrotoxicity** risk; use only short courses when clearly needed (e.g. severe sepsis, resistant organisms), with monitoring.
- **Nitrofurantoin near term**: neonatal haemolysis (especially in G6PD deficiency).

3. Acute pyelonephritis: admit every case

Step-by-step (HC IV and hospital)

1. **Assess ABCDE and screen for sepsis** (MEOWS; temperature, HR, RR, BP, SpO₂, mental state, urine output).
2. **IV access**; bloods: **FBC, creatinine and electrolytes, blood cultures, mRDT/smear, glucose, lactate** (where available), group and save if anaemic.
3. **Urine for culture before antibiotics** (MSU or catheter specimen).
4. **Start IV antibiotics immediately** (within 1 hour if septic):

Drug	Dose	Route	Frequency	Notes
Ceftriaxone (first line)	1 g (2 g if septic or severe)	IV (or IM)	Once daily	Continue until afebrile for 24–48 hours , then oral step-down
Ampicillin + gentamicin (alternative, e.g. where ceftriaxone unavailable)	Ampicillin 2 g 6-hourly + gentamicin 5 mg/kg once daily	IV	As stated	Short-course gentamicin; check renal function
Meropenem (ESBL or failure of ceftriaxone, per culture, referral hospital)	1 g	IV	8-hourly	Reserve drug; use culture guidance
Oral step-down	Cefixime 200 mg 12-hourly, or amoxicillin-clavulanate 625 mg 8-hourly, or cephalixin 500 mg 6–8-hourly (by sensitivity)	Oral		Complete a total of 10–14 days

5. **Fluids**: IV crystalloid (Ringer's lactate or 0.9% saline). If hypotensive: **500 mL bolus**, reassess, repeat to a total of about 30 mL/kg while watching for crackles and falling SpO₂. Otherwise maintenance fluids while vomiting. **Strict fluid balance chart**; aim for urine output **0.5 mL/kg/hour or more**. Stop fluids if she becomes breathless.
6. **Antipyretic and analgesia**: **paracetamol 1 g orally/IV 6-hourly** (max 4 g/day); fever increases fetal oxygen demand and uterine irritability. Avoid NSAIDs after 28 weeks.
7. **Antiemetic** if vomiting (e.g. metoclopramide 10 mg IV/IM 8-hourly).
8. **Treat co-existing malaria** if the test is positive (IV artesunate for severe malaria).

9. **Fetal assessment:** fetal heart rate (tachycardia usually settles with maternal fever), CTG if viable gestation; watch for contractions.
10. **Preterm labour:** give **corticosteroids** if 24–34 weeks and delivery is likely (dexamethasone 6 mg IM 12-hourly × 4). Contractions often settle once fever is controlled. **Avoid beta-agonist tocolytics** (salbutamol) because they increase the risk of **pulmonary oedema/ARDS**; if tocolysis is appropriate, use **nifedipine** and only when infection is being controlled.
11. **Thromboprophylaxis** (LMWH) if immobile, septic or with other risk factors.
12. **Monitor:** vital signs 1–4-hourly (MEOWS), fluid balance, SpO₂, daily FBC and creatinine in severe cases.
13. **Review at 48–72 hours:**
 - **Improving:** switch to oral when afebrile 24–48 hours and eating; discharge with the oral course.
 - **Not improving:** check culture/sensitivity, change antibiotics, **renal ultrasound** (obstruction, stone, abscess), reconsider diagnosis, involve urology (ureteric stent or percutaneous nephrostomy for infected obstruction).
14. **After treatment: test of cure** 1–2 weeks after, then **suppressive prophylaxis** for the rest of pregnancy (recurrence about 6–8%, higher without prophylaxis) and **monthly or each-trimester urine culture**.

UGANDA CONTEXT

At a HC III: recognise, give **the first dose of ceftriaxone 1 g IM/IV** (and antimalarial if mRDT positive), paracetamol, oral fluids, and **refer to the HC IV/hospital**. **At a HC IV:** admit, IV ceftriaxone, fluids, monitoring; **refer to the regional referral hospital** if septic shock, respiratory distress, AKI, no improvement in 48–72 hours, preterm labour below 34 weeks without neonatal care, or need for imaging/culture/urology.

4. Urosepsis and septic shock (algorithm)

1. **Recognise:** suspected infection + abnormal MEOWS (e.g. SBP 100 or below, RR 22 or more, altered mentation) or lactate above 2.
2. **Call for help** (senior obstetrician, anaesthetist/ICU).
3. **Within the first hour ("Sepsis Six"):**
 1. **Oxygen** to keep SpO₂ 94–98%.
 2. **Blood cultures** (and urine culture).
 3. **IV broad-spectrum antibiotics: ceftriaxone 2 g IV** (± gentamicin 5 mg/kg; meropenem if known ESBL or hospital-acquired).
 4. **IV fluids:** 500 mL crystalloid boluses, reassessing after each (up to about 30 mL/kg).
 5. **Lactate and Hb.**
 6. **Urine output:** catheterise, hourly measurement.
4. **Left lateral tilt** (relieve aortocaval compression) if more than 20 weeks.
5. **Persistent hypotension after fluids:** vasopressors (noradrenaline) in HDU/ICU.
6. **Source control:** relieve obstruction (stent/nephrostomy); drain abscess.
7. **Fetus:** CTG once the mother is stable; the fetal heart usually recovers with maternal resuscitation. **Deliver only for obstetric indications** or if intrauterine infection is the source.
8. **VTE prophylaxis,** glucose control, reassess every 30–60 minutes.

5. Recurrent UTI and prophylaxis

Indications: two or more UTIs in pregnancy, one UTI plus ASB, one episode of pyelonephritis (many units), or persistent bacteriuria after two courses.

Drug	Dose	Route	Frequency	Duration
Nitrofurantoin	50–100 mg	Oral	Once nightly (or after intercourse)	Until delivery (switch to cephalexin from about 36 weeks)
Cephalexin	250–500 mg	Oral	Once nightly	Until delivery or up to 6 weeks postpartum

Also:

- **Look for a cause:** diabetes (glucose), **sickle cell disease/trait**, HIV, **renal ultrasound** for stones/anomalies (and full imaging postpartum if recurrent), incomplete emptying.
- **Monthly urine culture** for the rest of the pregnancy.
- Behavioural advice: fluids, regular and post-coital voiding, front-to-back wiping.

6. Postpartum

- **Postpartum UTI** is common after catheterisation and CS; treat as above (nitrofurantoin is compatible with breastfeeding except for G6PD-deficient or jaundiced/preterm infants).
- Women with **recurrent pyelonephritis** or suspected structural disease need **renal imaging after 6–12 weeks postpartum** (when physiological dilatation has resolved).

Complications

Maternal	Fetal / neonatal
Pyelonephritis (from untreated ASB/cystitis)	Preterm birth
Sepsis and septic shock	Low birth weight
ARDS / pulmonary oedema (1–2% of pyelonephritis)	Fetal tachycardia, hypoxia, stillbirth in septic shock
Anaemia (haemolysis)	Neonatal sepsis (e.g. GBS)
Acute kidney injury (usually transient)	Drug effects (e.g. kernicterus with sulfonamides near term; haemolysis with nitrofurantoin at term in G6PD deficiency)
Perinephric/renal abscess, infected obstruction	Association with hypertensive disorders and growth restriction (ASB)
Recurrence (6–8%, higher without prophylaxis)	
Chronic pyelonephritis/renal scarring (rare, recurrent infection with reflux)	
Maternal death (sepsis)	

Ward-round & exam pearls

Model presentation

"Mrs B is a 24-year-old G2P1 at **26 weeks**, admitted last night with **2 days of fever, rigors, right loin pain and vomiting**, with a history of cystitis at 14 weeks that was not followed up. On admission: T 39.2 °C, HR 118, RR 22, BP 104/66, SpO₂ 97%, **right renal angle tenderness**, uterus soft and non-tender, no contractions, FHR 170. Dipstick: nitrites positive, LE 3+. **mRDT negative**. Hb 10.1 g/dL, WBC 17, creatinine 88

μmol/L. Impression: **acute right pyelonephritis with sepsis** (MEOWS triggers). She received **ceftriaxone 2 g IV** within an hour after blood and urine cultures, 1 L Ringer's lactate in boluses, and paracetamol. This morning: T 37.8 °C, HR 96, RR 18, urine output adequate, FHR 150, no contractions. **Plan:** continue ceftriaxone until afebrile 24–48 hours, then **oral cefixime to complete 14 days** guided by culture; fluid balance; watch SpO₂ for ARDS; CTG/contractions; **test of cure** in 1–2 weeks, then **nightly nitrofurantoin prophylaxis** and monthly urine cultures."

Likely viva questions

1. **Why is pregnancy a risk factor for pyelonephritis?** Progesterone-induced ureteric dilatation and hypoperistalsis, mechanical compression at the pelvic brim (right more than left), vesicoureteric reflux, glycosuria/aminoaciduria, larger bladder with residual urine.
2. **Why is the right side affected more?** Dextrorotation of the uterus, the sigmoid colon cushions the left ureter, and the right ovarian vein crosses the right ureter.
3. **Define ASB and explain why we treat it in pregnancy.** 10⁵ CFU/mL or more of a single organism without symptoms; 20–40% progress to pyelonephritis if untreated; treatment reduces pyelonephritis and low birth weight.
4. **Name antibiotics safe in pregnancy and those to avoid.** Safe: penicillins, cephalosporins, nitrofurantoin (not near term), fosfomycin. Avoid: fluoroquinolones, tetracyclines, trimethoprim (first trimester), sulfonamides (first trimester/term), prolonged aminoglycosides.
5. **Why not nitrofurantoin for pyelonephritis?** It concentrates in urine but achieves poor renal tissue and blood levels.
6. **Why avoid nitrofurantoin at term?** Risk of **haemolytic anaemia in the neonate** (immature glutathione system, G6PD deficiency).
7. **What are the complications of pyelonephritis in pregnancy?** Sepsis, septic shock, ARDS, anaemia, AKI, preterm labour, recurrence.
8. **What do you do if fever persists after 72 hours of IV ceftriaxone?** Review culture/sensitivity (ESBL?), renal ultrasound for obstruction/abscess, reconsider diagnosis (malaria, chorioamnionitis, appendicitis), change antibiotics, urology.
9. **What is a test of cure and when?** Repeat urine culture 1–2 weeks after treatment.
10. **What should be screened for in recurrent UTI?** Diabetes, sickle cell, HIV, stones, anomalies.

OSCE tips

- **Urinalysis station:** state the dipstick results systematically (leucocytes, nitrites, protein, blood, glucose, ketones, specific gravity, pH) and interpret each in pregnancy (e.g. protein = UTI or pre-eclampsia; glucose = physiological or diabetes; ketones = vomiting/starvation).
- **Prescribing station:** always state gestation-appropriate drug, dose, route, frequency, duration, test of cure, and safety-net advice.

Mnemonics & memory aids

Why pregnancy predisposes: "PROGRESS"

- **P**rogesterone relaxes smooth muscle
- **R**eflux (vesicoureteric)
- **O**bstruction by the gravid uterus (right more than left)
- **G**lycosuria and aminoaciduria
- **R**esidual urine in a larger bladder

- **E. coli** from the nearby perineum
- **Stasis** in dilated ureters
- **Suppressed immunity** (relative)

Antibiotics to avoid: "Stop The Fetal Toxic Antibiotics"

- **Sulfonamides** (first trimester and term)
- **Trimethoprim** (first trimester)
- **Fluoroquinolones**
- **Tetracyclines**
- **Aminoglycosides** (prolonged use)

Safe choices: "Nice Pregnant Cells Face Care": Nitrofurantoin (not at term), **Penicillins** (amoxicillin, co-amoxiclav), **Cephalosporins** (cephalexin, cefixime, ceftriaxone), **Fosfomycin**, **Clinical culture-guided therapy**.

Pyelonephritis management: "ADMIT": **A**ll cases admitted; **D**raw cultures (blood, urine) first; **M**edicare IV ceftriaxone; **I**ntake/output (careful fluids, watch for ARDS); **T**est of cure and prophylaxis afterwards.

High-yield summary

Numbers & doses to know

Item	Value
Significant bacteriuria (MSU)	10⁵ CFU/mL or more of one organism (lower on catheter specimen; GBS at any count)
ASB prevalence	2–10% (about 13% at Mulago)
Untreated ASB → pyelonephritis	20–40% (about 1–2% if treated)
Pyelonephritis	1–2% of pregnancies; mostly 2nd/3rd trimester; right more than left
<i>E. coli</i> share	70–80%
Dipstick nitrite sensitivity	Only about 50%
Microscopy pyuria	More than 10 pus cells/HPF ; white cell casts = pyelonephritis
Nitrofurantoin	100 mg MR 12-hourly (or 50–100 mg 6-hourly standard) × 5–7 days ; avoid from ~36–38 weeks, in labour, G6PD deficiency
Amoxicillin	500 mg 8-hourly × 7 days (only if sensitive)
Cephalexin	500 mg 8–12-hourly × 5–7 days
Amoxicillin-clavulanate / cefixime	625 mg 8-hourly / 200 mg 12-hourly , 5–7 days
ASB course (WHO)	7 days
Pyelonephritis IV	Ceftriaxone 1 g IV daily (2 g if septic) ; alternative ampicillin 2 g 6-hourly + gentamicin 5 mg/kg daily
IV to oral switch	Afebrile 24–48 h ; total course 10–14 days
Blood culture positive in pyelonephritis	15–20%
Sepsis triggers	T above 38 or below 36 , HR above 100 , RR 22 or more , SBP 100 or below , SpO ₂ below 95% , confusion, lactate above 2 (severe 4 or more)

Item	Value
Fluid bolus	500 mL crystalloid, reassess, up to about 30 mL/kg
Urine output target	0.5 mL/kg/h or more
Review for failure	48–72 h (culture, renal ultrasound)
Test of cure	1–2 weeks after treatment
Prophylaxis	Nitrofurantoin 50–100 mg nightly or cephalexin 250–500 mg nightly until delivery
GBS intrapartum prophylaxis	Benzylpenicillin 3 g IV , then 1.8 g IV 4-hourly ; clindamycin 900 mg IV 8-hourly if allergic
Recurrence after pyelonephritis	6–8% (higher without prophylaxis)

Classic exam traps

- **"Ciprofloxacin for UTI."** Wrong in pregnancy: **avoid fluoroquinolones**. Use nitrofurantoin, cephalexin, amoxicillin-clavulanate or ceftriaxone.
- **"Negative dipstick, so no ASB."** Wrong: nitrites are only ~50% sensitive; **culture** is the screen.
- **"Nitrofurantoin for pyelonephritis."** Wrong: poor renal tissue and blood levels. **IV ceftriaxone**.
- **"Nitrofurantoin at 39 weeks."** Wrong: neonatal haemolysis; use cephalexin near term.
- **"Treat pyelonephritis as an outpatient with oral antibiotics."** Wrong: **admit every case**.
- **"The mRDT is positive, so the fever is malaria."** Not necessarily: malaria and pyelonephritis co-exist, and HRP2 tests stay positive after treatment. Examine the loins and dip the urine.
- **"Give 3 L of fluid quickly because she is febrile."** Wrong: pregnant women with pyelonephritis are prone to **ARDS/pulmonary oedema**. **500 mL boluses** with reassessment.
- **"Salbutamol for the contractions."** Wrong: beta-agonists worsen pulmonary oedema. Treat the infection; use **nifedipine** if tocolysis is appropriate.

Questions seniors ask

1. **"How do you know this is pyelonephritis and not cystitis?"** Fever, rigors, loin pain and renal angle tenderness (upper tract). Cystitis has no fever or loin pain.
2. **"Is she septic?"** State the triggers: temperature, HR, RR, SBP, SpO₂, consciousness, urine output, lactate. One or more with infection = start the sepsis bundle within 1 hour.
3. **"Did you send a urine culture before the first dose?"** It should be sent before antibiotics; it is the only way to catch ESBL and to do a meaningful test of cure.
4. **"What antibiotic, what dose and for how long?"** Ceftriaxone 1 g IV daily (2 g if septic) until afebrile 24–48 hours, then oral cefixime, amoxicillin-clavulanate or cephalexin by sensitivity to complete 10–14 days.
5. **"She is still febrile at 72 hours. What now?"** Check culture (ESBL: meropenem), renal ultrasound for obstruction or abscess, rethink the diagnosis (malaria, chorioamnionitis, appendicitis, pneumonia), urology if obstructed.
6. **"Why is her Hb falling?"** Endotoxin-mediated **haemolysis** (about a quarter of women), plus dilution from IV fluids; also consider malaria.
7. **"What happens after discharge?"** Complete the oral course, **test of cure** in 1–2 weeks, **nightly prophylaxis** until delivery, monthly (or each-trimester) urine culture, look for a cause (diabetes, sickle cell, HIV, stones).
8. **"What does sterile pyuria make you think of?"** Chlamydia/gonorrhoea, recent antibiotics, contamination by vaginal discharge, and **urinary TB** (especially with HIV).

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Why does pregnancy predispose to pyelonephritis?	Progesterone relaxes ureteric smooth muscle (dilatation, reduced peristalsis, stasis); the gravid uterus compresses the ureters at the pelvic brim; vesicoureteric reflux ; glycosuria/aminoaciduria ; larger bladder with residual urine; relative immunosuppression.
2	Why is pyelonephritis in pregnancy more often right-sided?	The uterus is dextrorotated , the sigmoid colon cushions the left ureter , and the right ovarian vein crosses the right ureter, so the right ureter is compressed more.
3	Define asymptomatic bacteriuria.	10⁵ CFU/mL or more of a single uropathogen on a clean-catch midstream urine culture in a woman without urinary symptoms .
4	Why screen and treat ASB in pregnancy but not in most non-pregnant women?	Untreated, 20–40% of pregnant women with ASB develop pyelonephritis (versus about 1–2% if treated); treatment also reduces low birth weight/preterm birth .
5	Commonest organism causing UTI in pregnancy?	<i>Escherichia coli</i> (70–80%), then <i>Klebsiella</i> , <i>Proteus</i> , <i>Enterobacter</i> , <i>Enterococcus</i> , group B streptococcus, <i>Staph. saprophyticus</i> .
6	Recommended duration of treatment for ASB in pregnancy (WHO)?	A 7-day antibiotic regimen (5–7 days for cystitis), followed by a test of cure culture 1–2 weeks later.
7	Nitrofurantoin dose and main cautions in pregnancy?	100 mg MR orally 12-hourly (or 50–100 mg 6-hourly) for 5–7 days . Not for pyelonephritis ; avoid at term/in labour and in G6PD deficiency (haemolysis).
8	Three oral antibiotics safe for cystitis in pregnancy?	Nitrofurantoin, cephalexin 500 mg 8–12-hourly, amoxicillin 500 mg 8-hourly (if sensitive) or amoxicillin-clavulanate 625 mg 8-hourly ; also cefixime, fosfomycin 3 g single dose. 5–7 days.
9	Why avoid trimethoprim in the first trimester?	It is a folate antagonist (dihydrofolate reductase inhibitor), associated with neural tube and other congenital defects .
10	Why avoid sulfonamides near term?	They displace bilirubin from albumin , increasing the risk of neonatal jaundice and kernicterus (and haemolysis in G6PD deficiency).
11	Why avoid fluoroquinolones and tetracyclines in pregnancy?	Fluoroquinolones : cartilage/arthritis in animal studies. Tetracyclines : tooth discolouration , impaired fetal bone growth, maternal hepatotoxicity.
12	First-line inpatient treatment of pyelonephritis in pregnancy (Uganda)?	Admit ; urine and blood cultures, then IV ceftriaxone 1 g once daily (2 g if septic) until afebrile 24–48 hours, then oral step-down (cefixime, amoxicillin-clavulanate or cephalexin by sensitivity) to complete 10–14 days .
13	Why can't nitrofurantoin be used for pyelonephritis?	It concentrates in the urine but achieves poor renal tissue and blood levels , so it cannot treat parenchymal infection or bacteraemia.
14	Main complications of pyelonephritis in pregnancy?	Sepsis/septic shock, ARDS/pulmonary oedema, anaemia (haemolysis), AKI, preterm labour , abscess/obstruction, recurrence .
15	Why are pregnant women with pyelonephritis prone to ARDS?	Endotoxin-mediated capillary leak , plus pregnancy's low oncotic pressure and high plasma volume; worsened by aggressive IV fluids and beta-agonist tocolytics .

#	Question	Answer
16	What do you do if a woman is still febrile after 48–72 hours of IV ceftriaxone?	Check culture sensitivities (ESBL?), renal ultrasound (obstruction, stone, abscess), reconsider the diagnosis (malaria, chorioamnionitis, appendicitis, pneumonia), change antibiotics (e.g. meropenem per culture), involve urology.
17	Prophylaxis for recurrent UTI in pregnancy?	Nitrofurantoin 50–100 mg nightly (switch near term) or cephalexin 250–500 mg nightly , until delivery (or up to 6 weeks postpartum), with monthly urine cultures.
18	What should you do with group B streptococcus found in urine?	Treat it now (any colony count) and give intrapartum IV antibiotic prophylaxis (benzylpenicillin 3 g IV, then 1.8 g 4-hourly until delivery).
19	Internal vs external dysuria?	Internal (felt inside the urethra) suggests UTI ; external (urine stinging sore vulval skin) suggests vulvovaginitis (candida, trichomonas, herpes).
20	What does sterile pyuria suggest?	White cells without bacterial growth: chlamydia/gonorrhoea, urinary TB (think HIV), contamination with vaginal discharge, recent antibiotics, stones.
21	Key investigations for suspected pyelonephritis in a Ugandan ANC ward?	Urine dipstick and culture before antibiotics, blood cultures, FBC , creatinine, mRDT/blood smear , glucose, lactate where available, fetal heart/CTG; renal ultrasound if not improving.
22	What are the sepsis-screen triggers in a pregnant woman with suspected pyelonephritis?	Infection plus any of: temperature above 38 or below 36 °C , HR above 100 , RR 22 or more , SBP 100 mmHg or below , new confusion, SpO ₂ below 95% , oliguria (below 0.5 mL/kg/h) or lactate above 2 mmol/L . Start the sepsis bundle within 1 hour .
23	Booking MSU grows 10 ⁵ CFU/mL E. coli; dipstick was nitrite-negative and she has no symptoms. Diagnosis and plan?	Asymptomatic bacteriuria (the dipstick is only ~50% sensitive). Treat by sensitivity for 7 days (e.g. nitrofurantoin 100 mg MR 12-hourly), then a test-of-cure culture 1–2 weeks later , and rescreen each trimester.
24	Urine: LE +, nitrites negative, many squamous epithelial cells, mixed growth. What does this mean?	A contaminated sample , often from vaginal discharge. Examine for vulvovaginitis, treat the discharge, and repeat a proper clean-catch MSU before diagnosing UTI.
25	What urine microscopy finding points to pyelonephritis rather than cystitis?	White cell casts (formed in the renal tubules). Pyuria (more than 10 pus cells/HPF) and bacteria occur in both.
26	How should IV fluids be given to a pregnant woman with pyelonephritis and hypotension, and why?	500 mL crystalloid boluses , reassessing chest, SpO ₂ and BP after each (up to about 30 mL/kg), with a strict fluid balance chart and urine output 0.5 mL/kg/h or more . Endotoxin capillary leak plus low oncotic pressure make her prone to pulmonary oedema/ARDS .

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

- 1.** A 23-year-old G1 at 14 weeks has a routine booking urine culture growing 10^5 CFU/mL of *E. coli* sensitive to nitrofurantoin, cephalexin and amoxicillin-clavulanate. She has no urinary symptoms. What is the MOST appropriate management?
- A No treatment because she is asymptomatic
 - B Treat with nitrofurantoin (or cephalexin) for 7 days and repeat urine culture 1–2 weeks after treatment
 - C Ciprofloxacin 500 mg twice daily for 3 days
 - D Single-dose amoxicillin 3 g and no follow-up
 - E Admit for IV ceftriaxone
- 2.** A 28-year-old at 22 weeks presents with dysuria, frequency and suprapubic pain for 2 days. Temperature 36.9 °C, no loin tenderness. Dipstick: nitrites positive, leucocytes 2+. Which is the MOST appropriate treatment?
- A Doxycycline 100 mg twice daily for 7 days
 - B Ciprofloxacin 500 mg twice daily for 7 days
 - C Trimethoprim 200 mg twice daily for 3 days
 - D IV ceftriaxone 1 g daily for 14 days
 - E Nitrofurantoin 100 mg MR twice daily for 5–7 days after sending an MSU for culture
- 3.** A 25-year-old at 38 weeks has symptomatic cystitis. Her culture grows *E. coli* sensitive to nitrofurantoin, cephalexin and amoxicillin-clavulanate. Her previous baby needed phototherapy for 'G6PD-related jaundice'. What is the BEST choice? Advanced
- A Nitrofurantoin 100 mg MR twice daily for 7 days
 - B Co-trimoxazole 960 mg twice daily for 7 days
 - C Cephalexin 500 mg 8-hourly for 7 days
 - D Doxycycline 100 mg twice daily for 7 days
 - E No treatment until after delivery
- 4.** A 24-year-old at 26 weeks presents with fever 39.2 °C, rigors, right loin pain and vomiting for 2 days. HR 118, BP 104/66, RR 22. Renal angle tenderness on the right. Dipstick: nitrites positive. mRDT negative. What is the MOST appropriate management?
- A Admit; take urine and blood cultures, start IV ceftriaxone, careful IV fluids with fluid balance, paracetamol, and monitor mother and fetus
 - B Oral nitrofurantoin and review in 3 days
 - C Oral ciprofloxacin as an outpatient
 - D IV artesunate only
 - E Emergency caesarean section for maternal sepsis

5. A 27-year-old at 30 weeks has been on IV ceftriaxone for pyelonephritis for 72 hours and still spikes fevers to 39 °C with persistent left loin pain. Urine culture grows *Klebsiella* resistant to ceftriaxone (ESBL). What is the BEST next step? Advanced

- A Continue ceftriaxone for another week
- B Stop antibiotics and observe
- C Add oral nitrofurantoin
- D Change to an antibiotic guided by sensitivities (e.g. meropenem) and arrange a renal ultrasound to exclude obstruction or abscess
- E Start ciprofloxacin

6. A 26-year-old at 28 weeks is on day 2 of IV ceftriaxone and IV fluids (5 L in 24 hours) for pyelonephritis. She becomes breathless: RR 34, SpO₂ 86% on air, bilateral crackles. What is the MOST likely diagnosis? Advanced

- A Pulmonary embolism
- B Acute respiratory distress syndrome / pulmonary oedema complicating pyelonephritis
- C Asthma
- D Anxiety
- E Amniotic fluid embolism

7. A 22-year-old at 32 weeks with pyelonephritis develops regular painful contractions on day 1 of IV ceftriaxone. Cervix is 2 cm dilated. Temperature 38.8 °C. What is the MOST appropriate management? Advanced

- A Salbutamol infusion for tocolysis
- B Discharge with oral antibiotics
- C Immediate caesarean section
- D Stop antibiotics because they cause contractions
- E Paracetamol for fever, continue IV antibiotics, give corticosteroids (dexamethasone 6 mg IM 12-hourly × 4), avoid beta-agonists; consider nifedipine only if appropriate

8. A 29-year-old HIV-positive woman at 8 weeks on TLD and co-trimoxazole prophylaxis has symptomatic cystitis. Which antibiotic is MOST appropriate?

- A Trimethoprim 200 mg twice daily
- B Increase co-trimoxazole to 960 mg twice daily as treatment
- C Cephalexin 500 mg 8-hourly for 5–7 days (after sending culture)
- D Ciprofloxacin 500 mg twice daily
- E Doxycycline 100 mg twice daily

9. A 30-year-old at 24 weeks has had cystitis twice and pyelonephritis once in this pregnancy. She has just completed treatment and the test of cure is negative. What is the MOST appropriate next step?

- A Nightly nitrofurantoin 50–100 mg (or cephalexin 250–500 mg) until delivery, monthly urine cultures, and look for an underlying cause (glucose, sickle status, HIV, renal ultrasound)
- B No further action
- C Weekly ciprofloxacin
- D Elective caesarean section at 37 weeks
- E Stop all fluids at night to reduce urine output

10. A 21-year-old at 30 weeks presents with fever and right-sided abdominal pain. The HC IV midwife says 'it must be malaria' because the mRDT is positive. Examination shows right renal angle tenderness; urine has nitrites and leucocytes 3+. What is the BEST interpretation?

- A Malaria alone; treat with antimalarials only
- B Pyelonephritis alone; the mRDT must be false
- C Appendicitis; call the surgeons and give no antibiotics
- D Possible co-existing malaria and pyelonephritis: treat malaria and admit for IV antibiotics with urine and blood cultures
- E Normal pregnancy discomfort

11. A 26-year-old at 34 weeks with pyelonephritis is hypotensive (BP 82/48), HR 132, RR 28, confused, with lactate 4.6 mmol/L after the first litre of fluid. What is the MOST appropriate next step? Advanced

- A Continue oral antibiotics and observe on the ward
- B Septic shock: call senior/ICU help, oxygen, broad-spectrum IV antibiotics (e.g. ceftriaxone 2 g ± gentamicin), further 500 mL fluid boluses with reassessment, catheter, and vasopressors if hypotension persists
- C Immediate caesarean section
- D Give 4 L of IV fluid rapidly then reassess
- E Start tocolysis

12. A 24-year-old at 20 weeks has urine microscopy showing 30 pus cells per high-power field but repeated cultures show no growth. She has a vaginal discharge and her partner has urethral discharge. What is the MOST likely cause? Advanced

- A E. coli pyelonephritis
- B Group B streptococcal bacteriuria
- C Normal pregnancy finding
- D Renal cell carcinoma
- E Urethritis/cervicitis from chlamydia or gonorrhoea (sterile pyuria)

13. A 27-year-old at 36 weeks has group B streptococcus (10^4 CFU/mL) on urine culture. She is asymptomatic. What is the MOST appropriate plan?

- A No action because the count is below 10^5
- B Give intrapartum antibiotics only if she develops fever
- C Treat the bacteriuria now and give intrapartum IV benzylpenicillin prophylaxis in labour
- D Elective caesarean section
- E Treat the baby with antibiotics for 7 days after birth routinely

14. A 23-year-old at 12 weeks with suspected cystitis gives a urine sample. The lab reports mixed growth of three organisms with many squamous epithelial cells. What is the BEST next step?

- A Repeat a properly collected clean-catch midstream urine
- B Treat all three organisms with broad-spectrum antibiotics
- C Diagnose pyelonephritis
- D Perform suprapubic aspiration routinely
- E Ignore the symptoms

15. A 25-year-old at 30 weeks was treated for pyelonephritis with IV ceftriaxone for 3 days and is now afebrile for 48 hours and eating. What is the MOST appropriate discharge plan?

- A Stop antibiotics; she is cured
- B Oral ciprofloxacin for 7 days
- C Oral nitrofurantoin for 3 days only
- D Oral step-down (e.g. cefixime 200 mg 12-hourly) to complete 10–14 days total, test-of-cure culture 1–2 weeks later, then suppressive prophylaxis and regular cultures
- E IV ceftriaxone at home for 4 weeks

Answers & explanations

1. B · Treat with nitrofurantoin (or cephalexin) for 7 days and repeat urine culture 1–2 weeks after treatment

B is correct. ASB in pregnancy must be treated (20–40% progress to pyelonephritis if untreated), using a sensitive, pregnancy-safe antibiotic for **7 days** (WHO), then a **test of cure**. **A** ignores the risk. **C**: fluoroquinolones are avoided in pregnancy. **D**: single-dose regimens are less effective, and there is no test of cure. **E**: ASB is treated orally as an outpatient.

2. E · Nitrofurantoin 100 mg MR twice daily for 5–7 days after sending an MSU for culture

E is correct. Uncomplicated **cystitis** is treated orally with a pregnancy-safe agent such as **nitrofurantoin** (fine at 22 weeks), with culture before starting and a test of cure. **A**: tetracyclines stain teeth and affect bone. **C**: trimethoprim is best avoided (folate antagonist; mainly first trimester) and 3 days is short in pregnancy. **D**: IV therapy is for pyelonephritis. **B**: fluoroquinolones are avoided.

3. C · Cephalexin 500 mg 8-hourly for 7 days

C is correct. **Cephalexin** is safe at term. **A**: nitrofurantoin is avoided **at term** because of **neonatal haemolysis**, especially with a family history suggesting **G6PD deficiency**. **B**: sulfonamides near term risk **kernicterus** and haemolysis. **D**: tetracyclines are contraindicated. **E**: untreated cystitis may progress to pyelonephritis.

4. A · Admit; take urine and blood cultures, start IV ceftriaxone, careful IV fluids with fluid balance, paracetamol, and monitor mother and fetus

A is correct. This is **acute pyelonephritis with sepsis criteria**: admit, cultures first, **IV ceftriaxone** within an hour, careful fluids (ARDS risk), antipyretic and fetal monitoring. **B**: nitrofurantoin does not reach renal tissue; oral outpatient care is unsafe with vomiting and sepsis. **C**: fluoroquinolones are avoided and outpatient care is inappropriate. **D**: the mRDT is negative and there is a clear urinary source. **E**: delivery is not a treatment for pyelonephritis.

5. D · Change to an antibiotic guided by sensitivities (e.g. meropenem) and arrange a renal ultrasound to exclude obstruction or abscess

D is correct. Treatment failure at 48–72 hours needs **culture-guided antibiotics** (ESBL: a carbapenem such as meropenem) and **imaging** for obstruction, stones or abscess (which may need drainage/stenting). **A**: the organism is resistant. **C**: nitrofurantoin does not treat renal parenchyma. **B** is dangerous. **E**: fluoroquinolones are avoided in pregnancy, and ESBL organisms are often co-resistant.

6. B · Acute respiratory distress syndrome / pulmonary oedema complicating pyelonephritis

B is correct. Endotoxin-induced **capillary leak** plus pregnancy's low oncotic pressure and **large-volume IV fluids** cause **ARDS/pulmonary oedema** (1–2% of pyelonephritis). Stop fluids, oxygen, senior/ICU review, consider diuretic if overload. **A**: PE causes breathlessness but usually with a clear chest. **C**: no history, and wheeze rather than crackles. **D** would not cause hypoxia. **E** occurs in labour/delivery with collapse and coagulopathy.

7. E · Paracetamol for fever, continue IV antibiotics, give corticosteroids (dexamethasone 6 mg IM 12-hourly × 4), avoid beta-agonists; consider nifedipine only if appropriate

E is correct. Fever and inflammation trigger contractions, which often settle once fever is controlled. **Steroids** below 34 weeks improve neonatal outcome. **A:** beta-agonists increase the risk of **pulmonary oedema/ARDS** in pyelonephritis. **C:** no obstetric indication. **D** is wrong. **B:** she needs inpatient care.

8. C · Cephalixin 500 mg 8-hourly for 5–7 days (after sending culture)

C is correct. **Cephalixin** is safe in the first trimester and does not rely on the antifolate class to which her bacteria may already be exposed. **A:** trimethoprim is a folate antagonist to avoid in the **first trimester**. **B:** treatment-dose co-trimoxazole in the first trimester is best avoided, and bacteria in women on prophylaxis are often resistant. **D** and **E** are contraindicated in pregnancy.

9. A · Nightly nitrofurantoin 50–100 mg (or cephalixin 250–500 mg) until delivery, monthly urine cultures, and look for an underlying cause (glucose, sickle status, HIV, renal ultrasound)

A is correct. Recurrent UTI in pregnancy warrants **suppressive prophylaxis, monthly cultures** and a search for **risk factors** (diabetes, sickle cell, HIV, stones, anomalies). **B** leaves a high recurrence risk. **C:** fluoroquinolones are avoided. **D:** UTI is not an indication for CS. **E:** fluid restriction increases stasis and risk.

10. D · Possible co-existing malaria and pyelonephritis: treat malaria and admit for IV antibiotics with urine and blood cultures

D is correct. In Uganda **malaria and pyelonephritis can co-exist**, and an mRDT can remain positive after recent treated malaria; clear urinary findings with loin tenderness mean **pyelonephritis must be treated too**. **A** misses a sepsis risk. **B:** a positive mRDT should not be dismissed. **C:** appendicitis is a differential for right-sided pain, but urinary findings and renal angle tenderness favour pyelonephritis; antibiotics should not be withheld. **E** is unsafe.

11. B · Septic shock: call senior/ICU help, oxygen, broad-spectrum IV antibiotics (e.g. ceftriaxone 2 g ± gentamicin), further 500 mL fluid boluses with reassessment, catheter, and vasopressors if hypotension persists

B is correct. Hypotension despite fluids and lactate above 4 mean **septic shock**: follow the sepsis bundle with senior/ICU input and **vasopressors** if needed; left lateral tilt. **A** is inadequate. **C:** an unstable mother should be resuscitated first; delivery is for obstetric indications. **D:** large uncontrolled fluid volumes risk **ARDS**; give boluses with reassessment. **E** is irrelevant and harmful.

12. E · Urethritis/cervicitis from chlamydia or gonorrhoea (sterile pyuria)

E is correct. **Sterile pyuria** with discharge in her and her partner suggests an **STI** (chlamydia/gonorrhoea); manage with syndromic treatment safe in pregnancy (e.g. ceftriaxone + azithromycin) and partner treatment. **A:** would grow on culture and cause fever/loin pain. **C:** significant pyuria is not normal. **D** is very unlikely. **B:** GBS grows on culture.

13. C · Treat the bacteriuria now and give intrapartum IV benzylpenicillin prophylaxis in labour

C is correct. **GBS bacteriuria at any count** indicates heavy colonisation: treat now and give **intrapartum antibiotic prophylaxis** (benzylpenicillin 3 g IV then 1.8 g 4-hourly) to prevent early-onset neonatal GBS sepsis. **A:** the 10^5 threshold does not apply to GBS. **B** is too late for prevention. **D:** CS does not replace prophylaxis and is not indicated. **E:** routine neonatal antibiotics are not recommended if the mother received adequate prophylaxis.

14. A · Repeat a properly collected clean-catch midstream urine

A is correct. Mixed growth with epithelial cells indicates **contamination**; teach clean-catch technique (separate labia, clean front to back, midstream sample, lab within 2 hours) and repeat. If symptomatic, empirical treatment can start after the repeat sample is sent. **B** treats contamination. **C**: no clinical features of pyelonephritis. **D**: invasive and unnecessary. **E**: symptoms still need assessment.

15. D · Oral step-down (e.g. cefixime 200 mg 12-hourly) to complete 10–14 days total, test-of-cure culture 1–2 weeks later, then suppressive prophylaxis and regular cultures

D is correct. Complete **10–14 days** with an oral agent to which the organism is sensitive, then **test of cure, prophylaxis** (recurrence 6–8%) and regular cultures. **A**: 3 days is inadequate for pyelonephritis. **C**: nitrofurantoin does not treat renal parenchyma and 3 days is too short. **B**: fluoroquinolones are avoided. **E**: unnecessary and impractical.

Clinical cases

Work through each step before reading the model answer.

Case 1: Asymptomatic bacteriuria at booking

Harriet, 20, G1 at 13 weeks, books at a HC IV in Mbale. She is well. The booking urine dipstick shows nitrites positive and leucocytes 1+. The hospital lab 30 km away can do cultures.

1. What is the likely diagnosis, and why does it matter in pregnancy?

Asymptomatic bacteriuria (confirm by culture: 10^5 CFU/mL or more of one organism). It matters because pregnancy's ureteric dilatation and stasis mean **20–40% progress to pyelonephritis** if untreated, with risks of sepsis and preterm birth; treatment reduces pyelonephritis and low birth weight.

2. What history and checks do you add?

Symptoms (internal dysuria, frequency beyond her usual, fever, loin pain), previous UTIs, diabetes risk factors, **sickle status**, HIV status, drug allergies, G6PD history, vaginal discharge. Check BP, temperature, urine protein and glucose. Teach **clean-catch MSU** technique and send to the lab within 2 hours (or refrigerate).

3. Culture grows *E. coli* 10^5 CFU/mL, resistant to amoxicillin and co-trimoxazole, sensitive to nitrofurantoin and cephalexin. What do you prescribe?

Nitrofurantoin 100 mg MR orally 12-hourly for 7 days (or 50–100 mg 6-hourly), or **cephalexin 500 mg 8-hourly for 7 days**. Not amoxicillin (resistant), not co-trimoxazole or trimethoprim (resistant; first trimester), never ciprofloxacin or doxycycline. Advise fluids, regular voiding, and return if fever or loin pain.

4. What follow-up does she need?

Test-of-cure culture 1–2 weeks after finishing, then culture **each trimester** (or monthly if recurrent). If ASB recurs, re-treat by sensitivities and consider **nightly prophylaxis** until delivery. Look for causes if recurrent (glucose, sickle, HIV, renal ultrasound).

Take-away: Screen at booking, confirm with culture, treat ASB for 7 days with a pregnancy-safe drug chosen by sensitivity, and always do a test of cure.

Case 2: Pyelonephritis with sepsis at 26 weeks

Night admission at Mengo: Proscovia, 24, G2P1 at 26 weeks, with 2 days of fever, rigors, right loin pain and vomiting. T 39.2 °C, HR 118, RR 22, BP 104/66, SpO₂ 97%. Right renal angle tenderness. Uterus soft, no contractions, FHR 170.

1. Summarise the likely diagnosis and immediate differentials.

Acute right pyelonephritis with sepsis (MEOWS triggers). Differentials: **malaria** (can co-exist), appendicitis (right-sided, higher in pregnancy), chorioamnionitis (uterus non-tender), pneumonia, cholecystitis, renal stone with infected obstruction.

2. What do you do in the first hour?

Senior informed; **IV access**; bloods: FBC, creatinine, electrolytes, **blood cultures**, **mRDT**, glucose, lactate; **urine for culture before antibiotics**; **IV ceftriaxone 2 g** within the hour; **Ringer's lactate 500 mL bolus** and reassess; **paracetamol 1 g**; antiemetic; catheter or strict intake/output; oxygen if SpO₂ falls; left lateral position; FHR monitoring.

3. What do you monitor over the next 48 hours, and which complications do you look for?

MEOWS 1–4-hourly, **SpO₂ and RR** (ARDS/pulmonary oedema, especially with fluids), urine output (AKI), temperature curve, Hb (**haemolytic anaemia**), creatinine, **contractions** (preterm labour; steroids if delivery likely, avoid salbutamol), fetal heart (tachycardia should settle as fever falls).

4. On day 3 she is still febrile at 38.9 °C. Culture shows E. coli resistant to ceftriaxone (ESBL). What now?

Switch to a **culture-guided agent** (e.g. **meropenem 1 g IV 8-hourly**), request **renal ultrasound** (obstruction, stone, abscess; hydronephrosis out of proportion to gestation), involve urology if obstructed (stent/nephrostomy), recheck malaria and other sources.

5. She improves and is afebrile for 48 hours. How do you discharge her?

Oral step-down according to sensitivities to complete **10–14 days** (for an ESBL organism, options are limited: nitrofurantoin is not adequate for pyelonephritis; discuss with microbiology, e.g. complete IV therapy or use an oral agent to which it is sensitive and which is safe in pregnancy). **Test of cure** 1–2 weeks later, **nightly prophylaxis** (e.g. nitrofurantoin 50–100 mg if sensitive) until delivery, **monthly cultures**, safety-net advice.

Take-away: Pyelonephritis in pregnancy is a sepsis emergency: cultures first, IV antibiotics within the hour, careful fluids, watch for ARDS and preterm labour, and escalate early if not better at 48–72 hours.

Case 3: Recurrent UTIs and an underlying cause

Sylvia, 30, G4P3 at 28 weeks, attends ANC at a regional referral hospital with her third symptomatic UTI of this pregnancy. She is overweight and her last baby weighed 4.2 kg. She has never had an HIV test in this pregnancy.

1. What underlying causes should you look for, and why?

Diabetes/GDM (glycosuria feeds bacteria; previous macrosomia and obesity are risk factors), **HIV** (immunosuppression), **sickle cell disease/trait** (papillary damage, concentrating defect), **urinary stones or anomalies**, incomplete bladder emptying, sexual/hygiene factors, and inadequate treatment (resistant organism, no test of cure).

2. Which investigations do you order?

MSU culture and sensitivity; **75 g OGTT** (she is 28 weeks with risk factors); **HIV test**; **Hb electrophoresis/sickling test** if status unknown; creatinine; **renal ultrasound** (stones, hydronephrosis out of proportion, anomalies).

3. Culture grows Proteus sensitive to cephalixin; OGTT confirms GDM; renal ultrasound shows a right renal pelvis stone. How do you manage her?

Treat the UTI: **cephalexin 500 mg 8-hourly for 7 days**; test of cure. Start **prophylaxis: cephalexin 250–500 mg nightly** until delivery. Manage **GDM** (diet, monitoring, metformin/insulin as needed). **Stone:** urology review; Proteus (urease-producing) favours struvite stones; stent/nephrostomy only if obstruction or infection fails to settle; definitive treatment after delivery.

4. What follow-up is needed after delivery?

Continue prophylaxis up to 6 weeks postpartum if advised; **postpartum GDM testing at 6–13 weeks; renal imaging** after physiological dilatation resolves and stone management; counsel about early urine screening in future pregnancies.

Take-away: Recurrent UTI in pregnancy is a clue: treat and prevent (nightly prophylaxis), then hunt for the cause (diabetes, HIV, sickle cell, stones, anomalies).

References

1. Uganda Clinical Guidelines 2023: National Guidelines for Management of Common Health Conditions (urinary tract infections; infections in pregnancy). *Ministry of Health Uganda*, 2023. <https://www.differentiatedservicedelivery.org/wp-content/uploads/UCG-2023-Publication-Final-PDF-Version-1.pdf>
2. WHO recommendations on antenatal care for a positive pregnancy experience (asymptomatic bacteriuria screening and 7-day treatment). *World Health Organization*, 2016.
3. Antibiotics for asymptomatic bacteriuria in pregnancy (Cochrane Review CD000490). *Smaill FM, Vazquez JC. Cochrane Database of Systematic Reviews*, 2019.
4. Clinical Practice Guideline for the Management of Asymptomatic Bacteriuria: 2019 Update by the Infectious Diseases Society of America. *Nicolle LE, et al. Clinical Infectious Diseases*, 2019.
5. Microbial aetiology and sensitivity of asymptomatic bacteriuria among ante-natal mothers in Mulago hospital, Uganda. *Andabati G, Byamugisha J. African Health Sciences*, 2010.
6. Sulfonamides, Nitrofurantoin, and Risk of Birth Defects (Committee Opinion No. 717). *American College of Obstetricians and Gynecologists*, 2017.
7. Prevention of Early-onset Neonatal Group B Streptococcal Disease (Green-top Guideline No. 36). *Royal College of Obstetricians and Gynaecologists*, 2017.
8. Williams Obstetrics, 26th edition: Renal and Urinary Tract Disorders chapter. *McGraw Hill*, 2022.

A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.